

Object Oriented Programming (OOP) in Python

Learn:

- What is OOP?
- Why do we use OOP?
- Class and Object
- Attributes and Methods

What is OOP?

- OOP (Object-Oriented Programming) is a way of writing programs using Classes and Objects.
- Everything around us can be considered an object.

Real Life Example

Think about a Car.

Car has:

- Color
- Brand
- Speed

Car can:

- Start
- Stop
- Move

In OOP:

- Data → Attributes
- Actions → Methods

Class

- A Class is a blueprint or design.

Example

Before building a house, we create a map.

House Map = Class

Actual House = Object

```
class Student:
```

```
    pass
```

Object

- An Object is a real thing created from a class.

Example:

Class = Student

Objects:

Rahul

Ayesha

Aman

```
class Student:
```

```
    pass
```

```
s1 = Student()
```

Attributes

- Attributes store information about an object.

Example:

Student has:

Name

Age

Methods(Functions)

- **Methods are functions inside a class.**

```
class Student:
```

```
    name = "Rahul"
```

```
    def study(self):
```

```
        print("Studying")
```

```
s1 = Student()
```

```
print(s1.name)
```

```
s1.study()
```